

GEOSPATIAL AI WORKSPACE

MUSTATIL 6

Illustrated User Guide

Detection, Satellite, Training, Pipelines, GIS, SAM2 and Geospatial Tools

Step by step

Original screenshots

English / original UI labels

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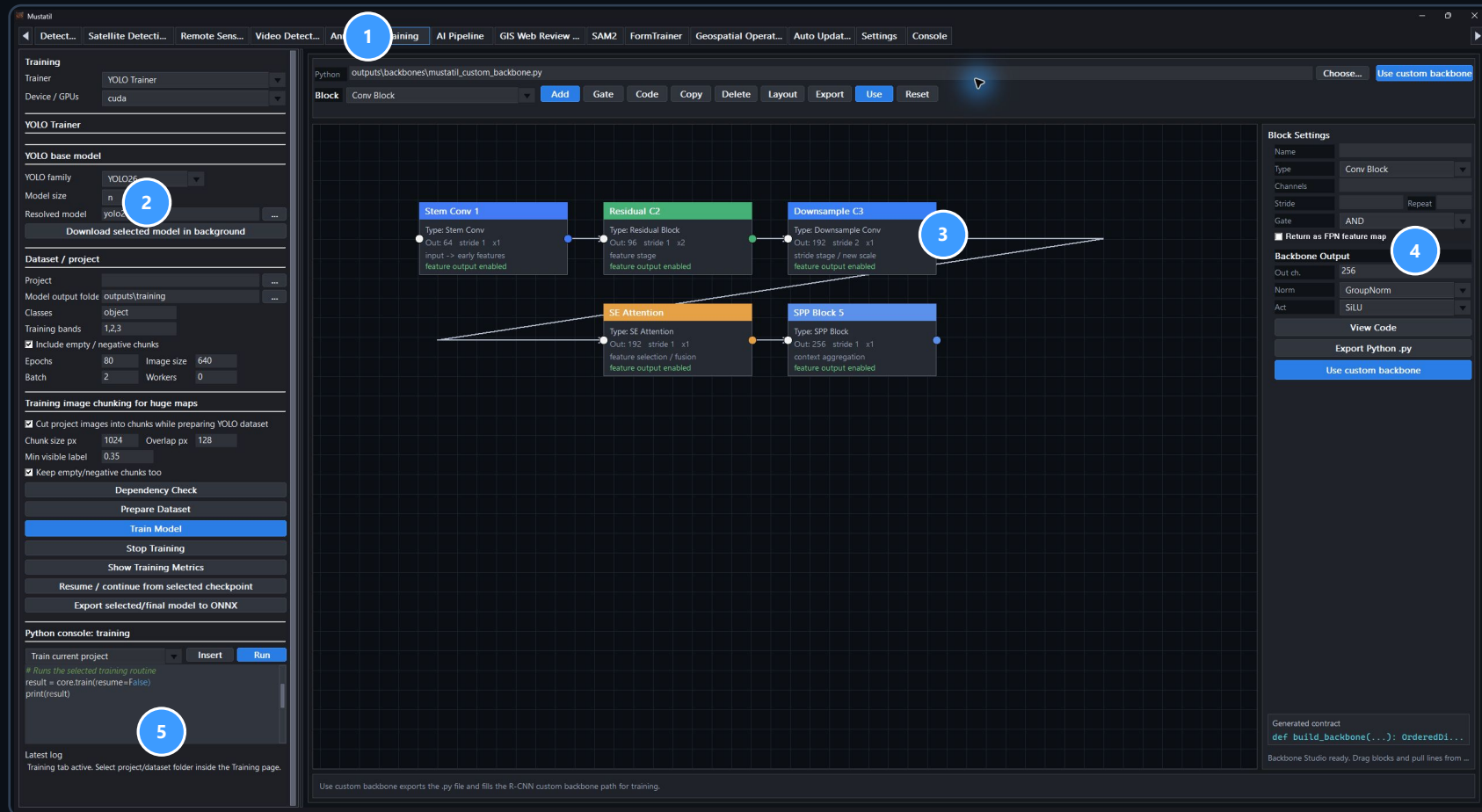
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Work safely

This guide does not start training runs, downloads, or exports automatically. Before starting, always check the model, device, input, output folder, and available storage. Use the Stop button in the active tab before closing Mustatil.

The Interface at a Glance



1 Main tabs

Switch between Detection, Training, GIS, and the other workspaces.

2 Settings

Inputs, models, devices, and project paths are usually configured on the left.

3 Workspace

The map, image, video, graph, or preview appears in the center.

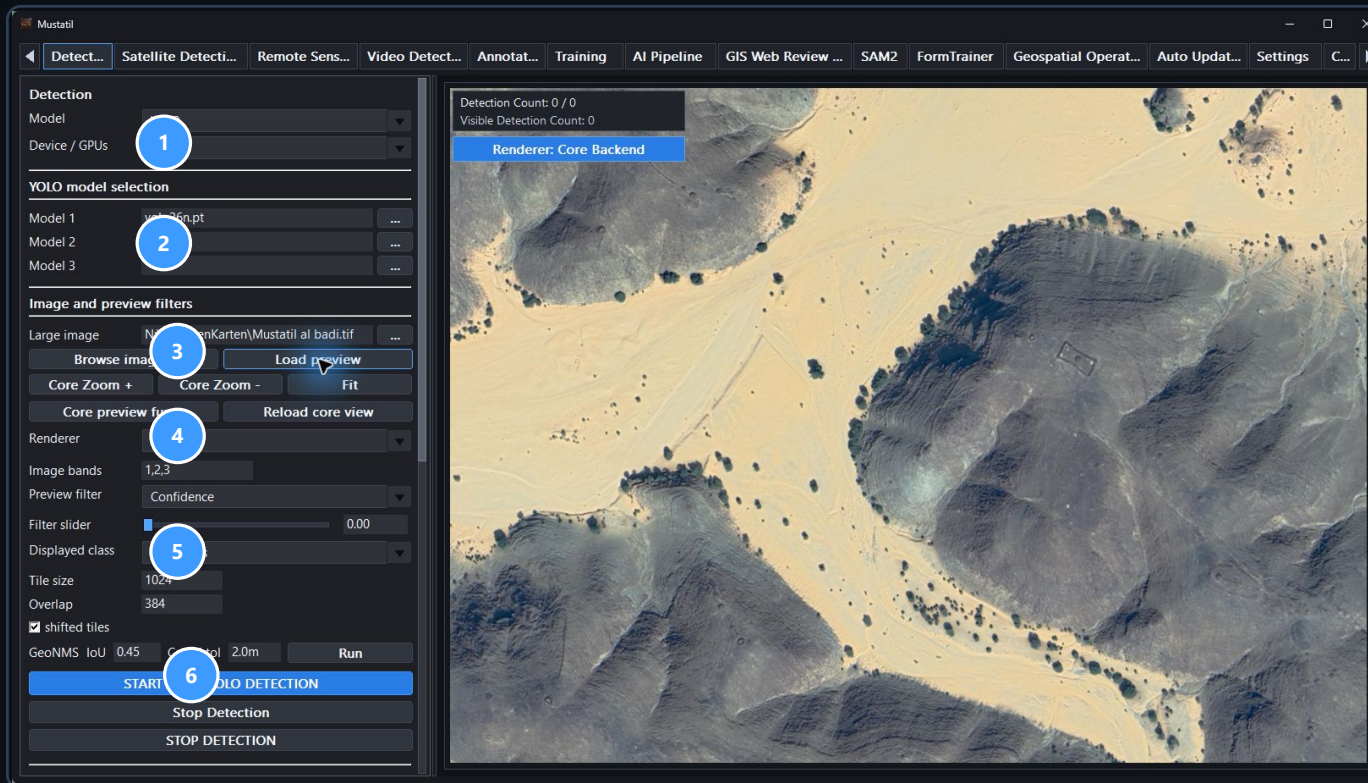
4 Properties

Configure selected blocks or objects on the right.

5 Log / Console

Status messages, Python templates, and errors appear at the bottom.

Quick Start: Detect Objects in a Large Raster



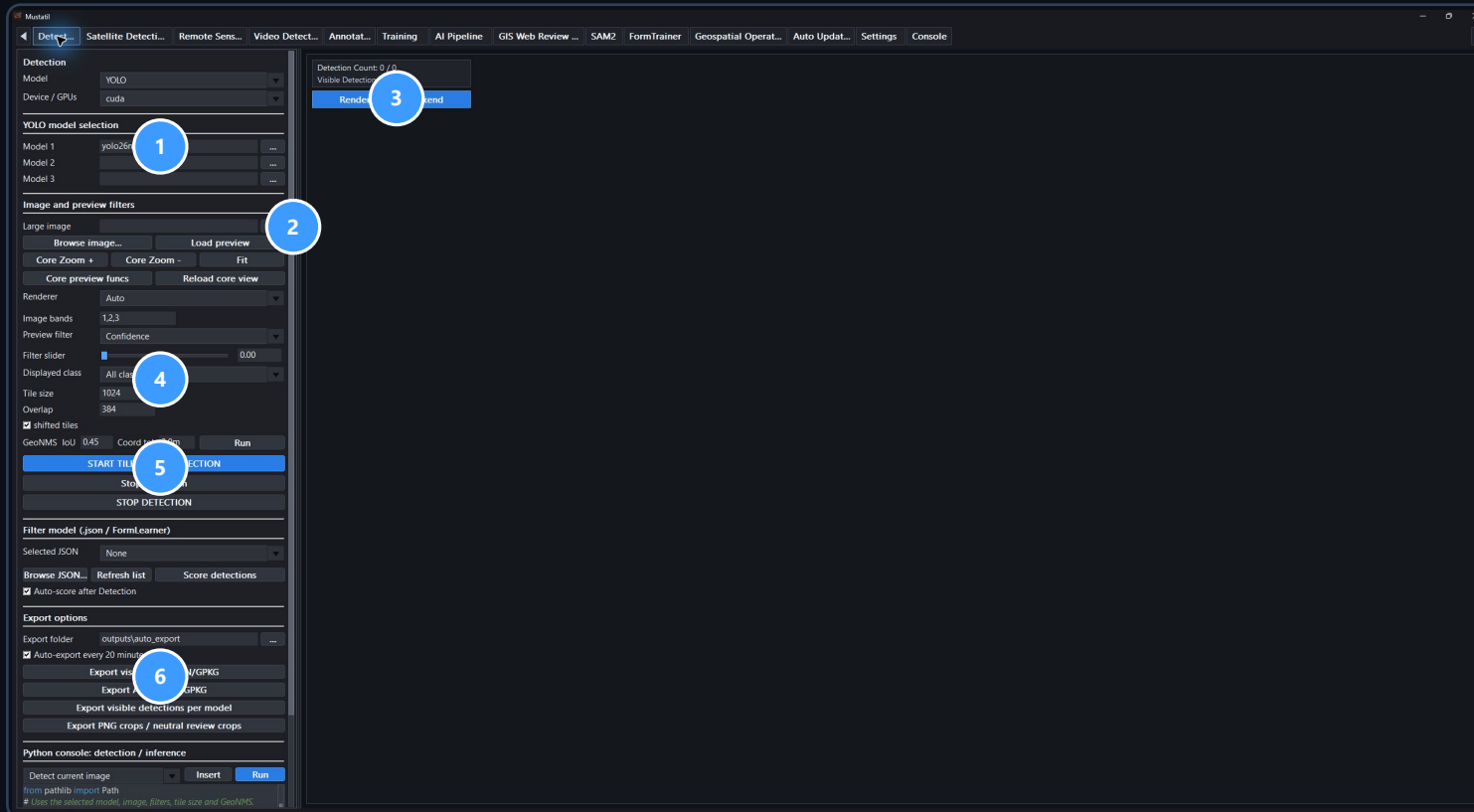
FROM IMAGE TO GPKG

- 1 Select the model backend at the top and choose cuda under Device / GPUs whenever possible.
- 2 Select at least Model 1 under YOLO model selection.
- 3 Click Browse image..., select the raster, and then click Load preview.
- 4 Choose the renderer, bands, confidence, tile size, overlap, and GeoNMS settings.
- 5 Click START TILED YOLO DETECTION. Raster prefetching and inference run in parallel.
- 6 Review the visible boxes, then export GeoJSON/GPKG or the model results.

For very large maps

Enable Auto-export and choose a writable local output folder. Stop Detection ends the active job safely.

Detection: Renderer, Tiling, and Export



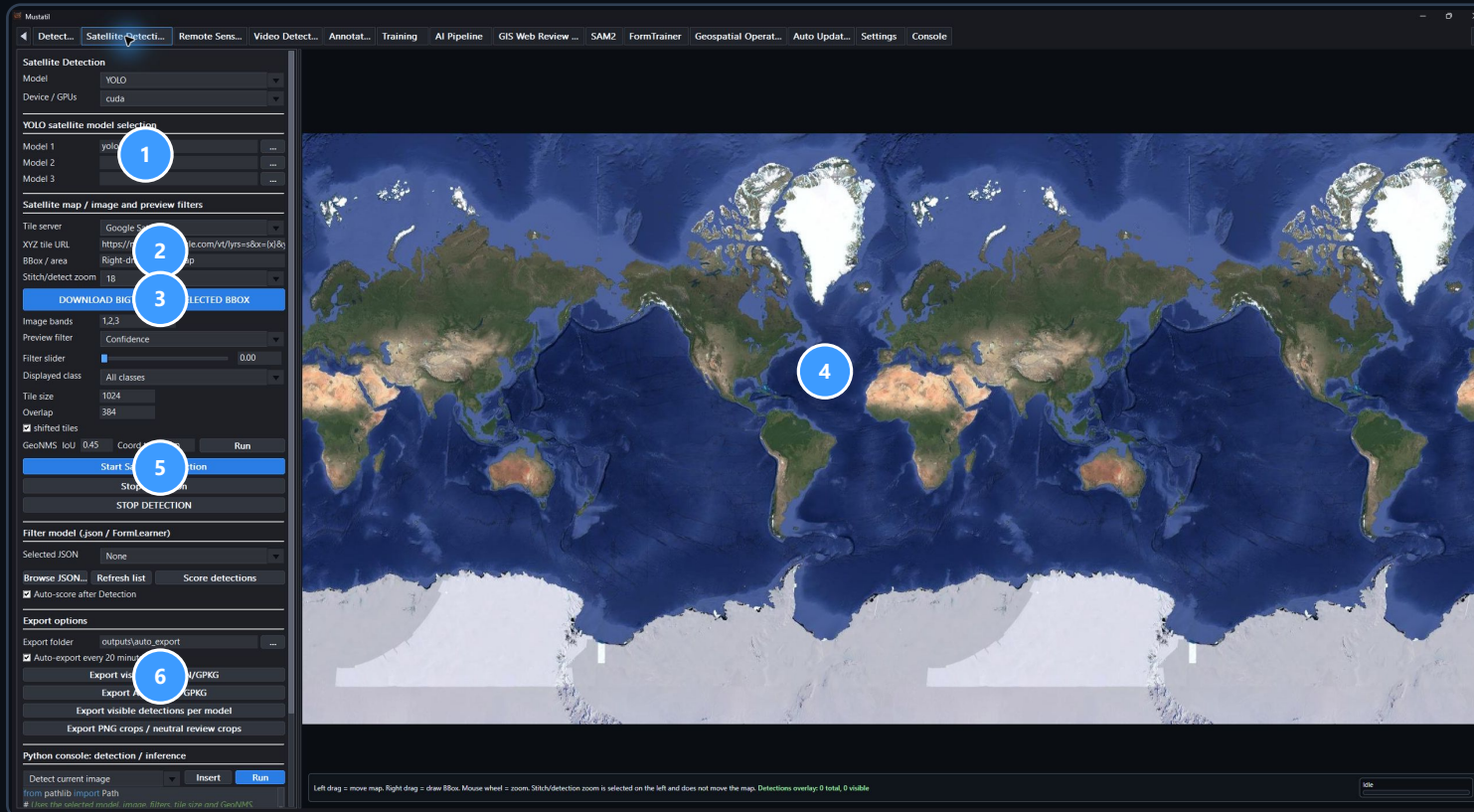
STEP BY STEP

- 1 Model 1-3 lets you use several models; their results can be exported separately.
- 2 Select the raster under Large image. Load preview prepares the display.
- 3 The blue renderer button switches between Core Backend and GDAL/Microtiles.
- 4 Filter slider and Displayed class change only the visible result display.
- 5 Tile size, Overlap, and shifted tiles control tiling; GeoNMS merges overlapping detections.
- 6 Export visible GeoJSON/GPKG saves the current view; Export ALL info to GPKG saves the complete run.

CUDA check

Choose Device / GPUs = cuda and verify startup in the log. High CPU usage during Rasterio prefetching is normal, but inference should still run on the GPU.

Satellite Detection: Select and Process an Area



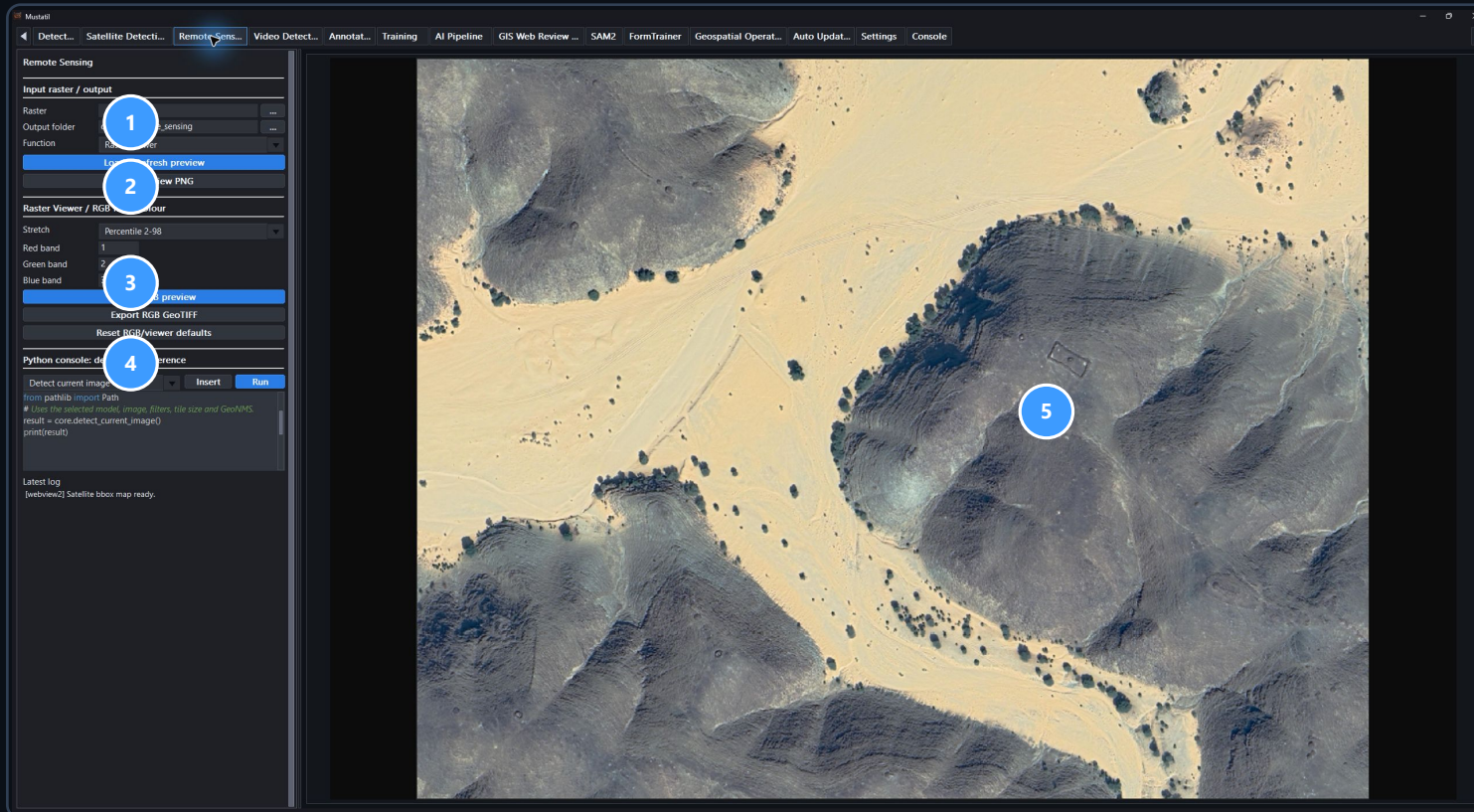
STEP BY STEP

- 1 Select the model and Device / GPUs; add up to three YOLO models if required.
- 2 Select the tile server and verify the XYZ tile URL.
- 3 Pan by left-dragging and zoom with the mouse wheel. Right-drag to draw the required area.
- 4 Set Stitch/detect zoom. Higher values provide more detail but require more tiles.
- 5 Use DOWNLOAD BIGTIFF FROM SELECTED BBOX or Start Satellite Detection. Downloading and detection can proceed in parallel.
- 6 Stop Detection ends processing; the export buttons save either visible or complete results.

Tip

Select the area roughly first, zoom in, and draw the BBox again. This avoids unnecessary downloads outside the target area.

Remote Sensing: Inspect and Display Rasters



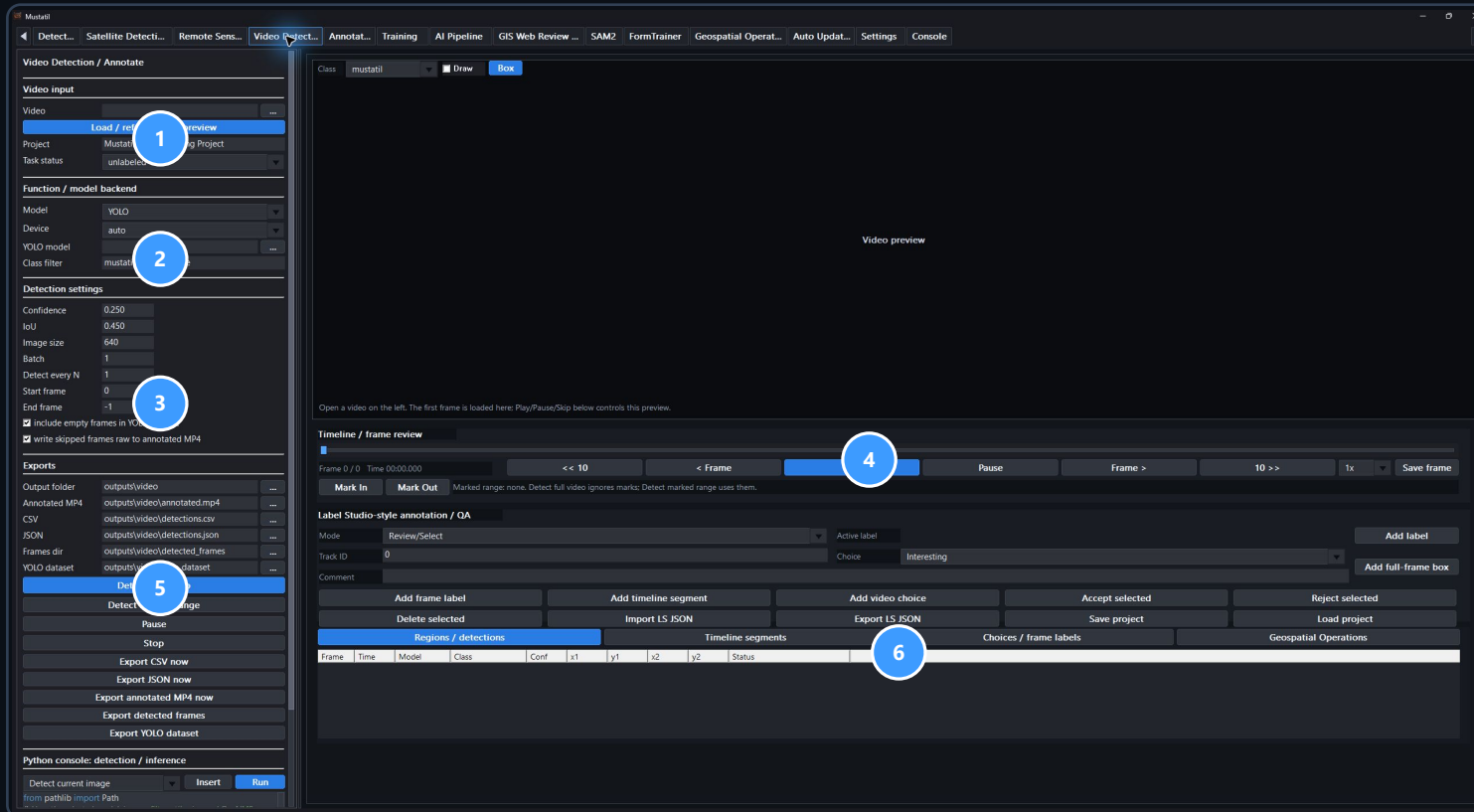
STEP BY STEP

- 1 Select the input file under Raster and choose an Output folder.
- 2 Choose the Function. Use Raster Viewer for visual inspection only.
- 3 Click Load / Refresh preview to load the raster through Rasterio/GDAL.
- 4 Adjust Stretch and the Red, Green, and Blue bands, then click Apply RGB preview.
- 5 Check the result in the large preview. Export RGB GeoTIFF saves the current band combination.

Note

Percentile 2-98 is a robust starting point for low-contrast satellite images. For multiband data, the band numbers must match the file.

Video Detection and Frame Review



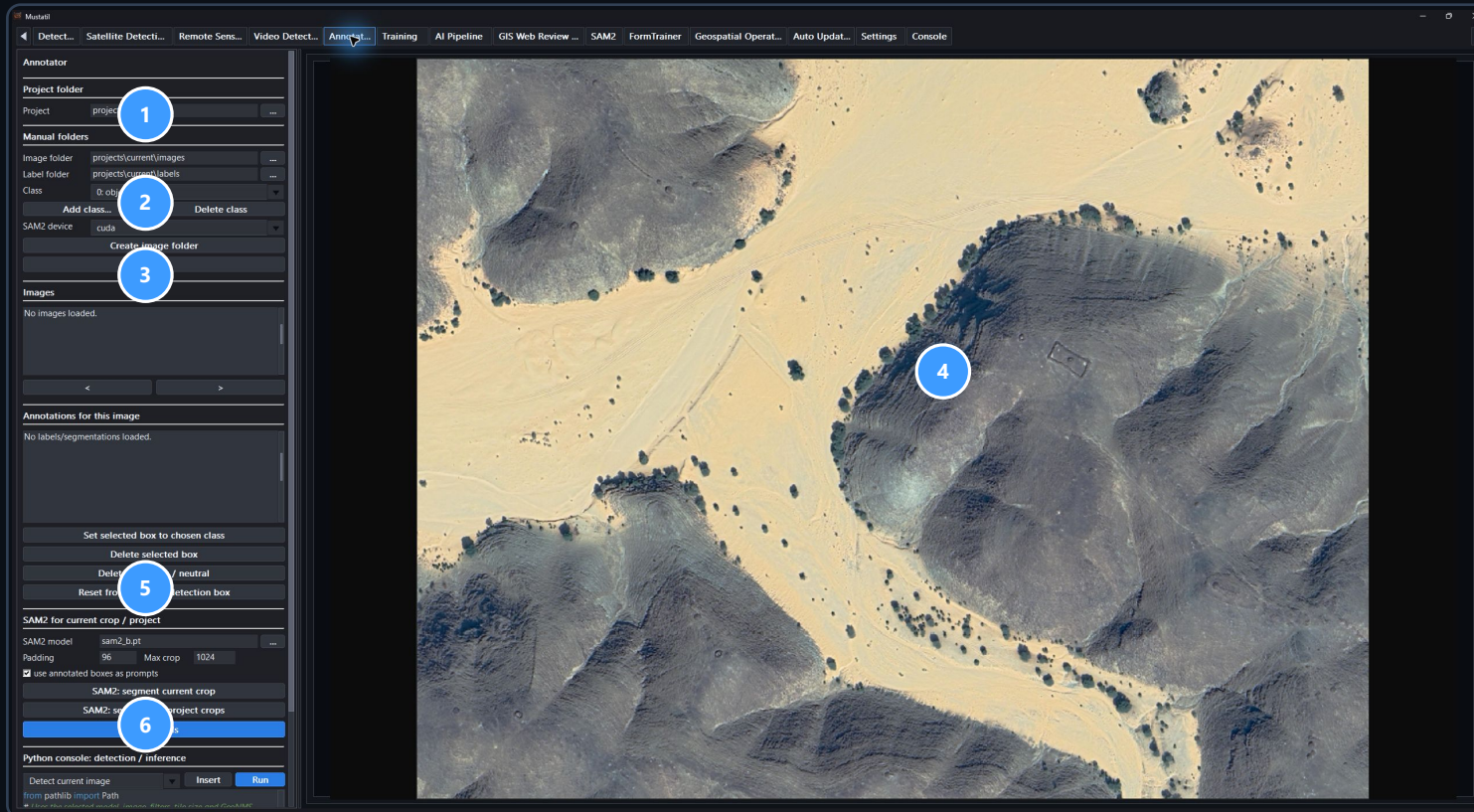
STEP BY STEP

- 1 Select the video and click Load / refresh video preview.
- 2 Set Model, Device, YOLO model, and optionally Class filter.
- 3 Configure Confidence, IoU, Image size, Batch, Detect every N, and the Start/End frame.
- 4 Use Timeline, Play/Pause, and Mark In/Mark Out to review a section.
- 5 Set the Output folder and export paths, then start Detect full video or Detect marked range.
- 6 Accept or reject detections in the QA area, then export CSV, JSON, MP4, frames, or a YOLO dataset.

Performance

Detect every N > 1 skips intermediate frames and speeds up long videos. Keep the value low for fast motion.

Annotation: Boxes, Classes, and SAM2 Prompts



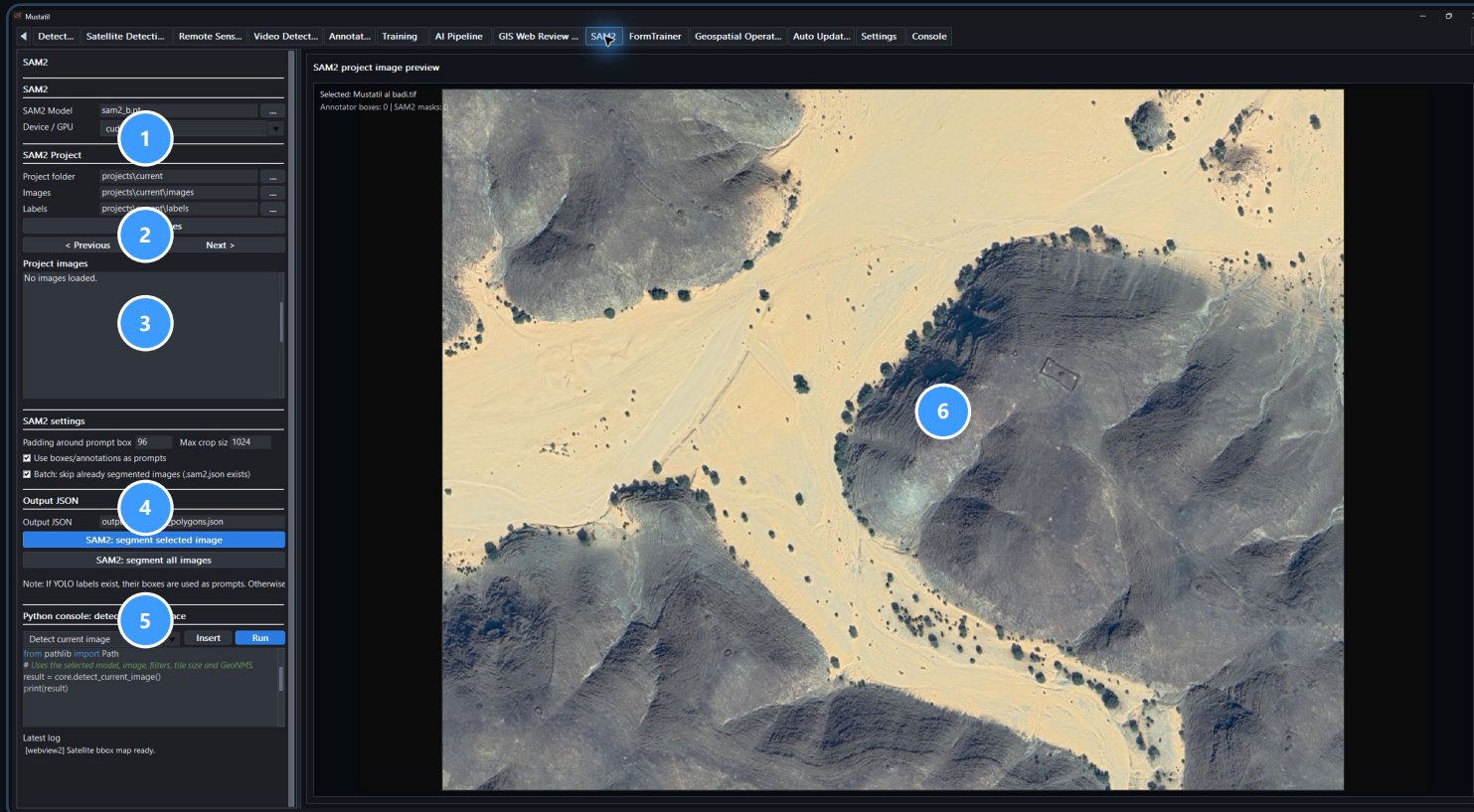
STEP BY STEP

- 1 Set the Project, Image folder, and Label folder.
- 2 Create or select a Class. The class format follows YOLO labels.
- 3 Click Create image folder and Load, then select an image from the list.
- 4 Draw boxes with the left mouse button. Zoom with the wheel; pan with the middle or right mouse button.
- 5 Assign the selected box to a class, delete it if needed, or restore it from the Detection manifest.
- 6 Use SAM2 for the current crop or the entire project, then finish with Save labels.

Safety

Delete all boxes / neutral removes every box from the current image. Use this button only after saving a backup.

SAM2: Segmentation from Box Prompts



STEP BY STEP

- 1 Select the SAM2 Model and Device / GPU. Choose cuda for GPU operation.
- 2 Set Project folder, Images, and Labels for the required project.
- 3 Click Load images and select an image with Previous/Next or from the list.
- 4 Enable Use boxes/annotations as prompts and adjust Padding and Max crop size.
- 5 Choose Output JSON and start SAM2: segment selected image or SAM2: segment all images.
- 6 Review masks and polygon outlines in the preview; existing .sam2.json files can be skipped during batch processing.

Effective prompts

Clean, tightly fitted boxes usually produce better masks. Excessive padding may include neighboring objects in the crop.

Training: Prepare the Dataset and Start a Model

The screenshot displays the Mustatil 6 Training interface, which is divided into several panels. On the left, the 'Training' panel contains settings for the YOLO Trainer, including the YOLO family, model size, and resolved model. Below this, the 'Dataset / project' panel allows for setting the project folder, classes, and training bands. The 'Training image chunking' panel includes options for cutting project images into chunks, setting chunk size and overlap, and keeping empty/negative chunks. The 'Dependency Check' panel shows the results of a dependency check. The 'Python console: training' panel provides a command to run the training process. The main central area is a 'Block' canvas where a custom backbone is being built using various blocks like Stem Conv, Residual, Downsample, SE Attention, and SPP. The right panel shows 'Block Settings' and 'Backbone Output' details. The bottom of the interface includes a 'Latest log' and a 'Use custom backbone' button.

STEP BY STEP

- 1 Select the Trainer and Device / GPUs. cuda uses a GPU; a multi-GPU selection appears in the field when supported.
- 2 Choose the model family, model size, and optionally the pretrained base model.
- 3 Choose folders for Project and Model output folder, then enter Classes and Training bands.
- 4 Set Epochs, Image size, Batch, and Workers to suit GPU memory and storage performance.
- 5 Enable chunking for large maps and choose Chunk size, Overlap, and Min visible label.
- 6 Run Dependency Check, then click Prepare Dataset.
- 7 Start Train Model. Use Show Training Metrics, Stop Training, Resume, or Export selected/final model to ONNX.

Workflow

Preparation converts the dataset into training tiles and labels. Start Train Model only after preparation has completed successfully.

Custom Backbone Studio

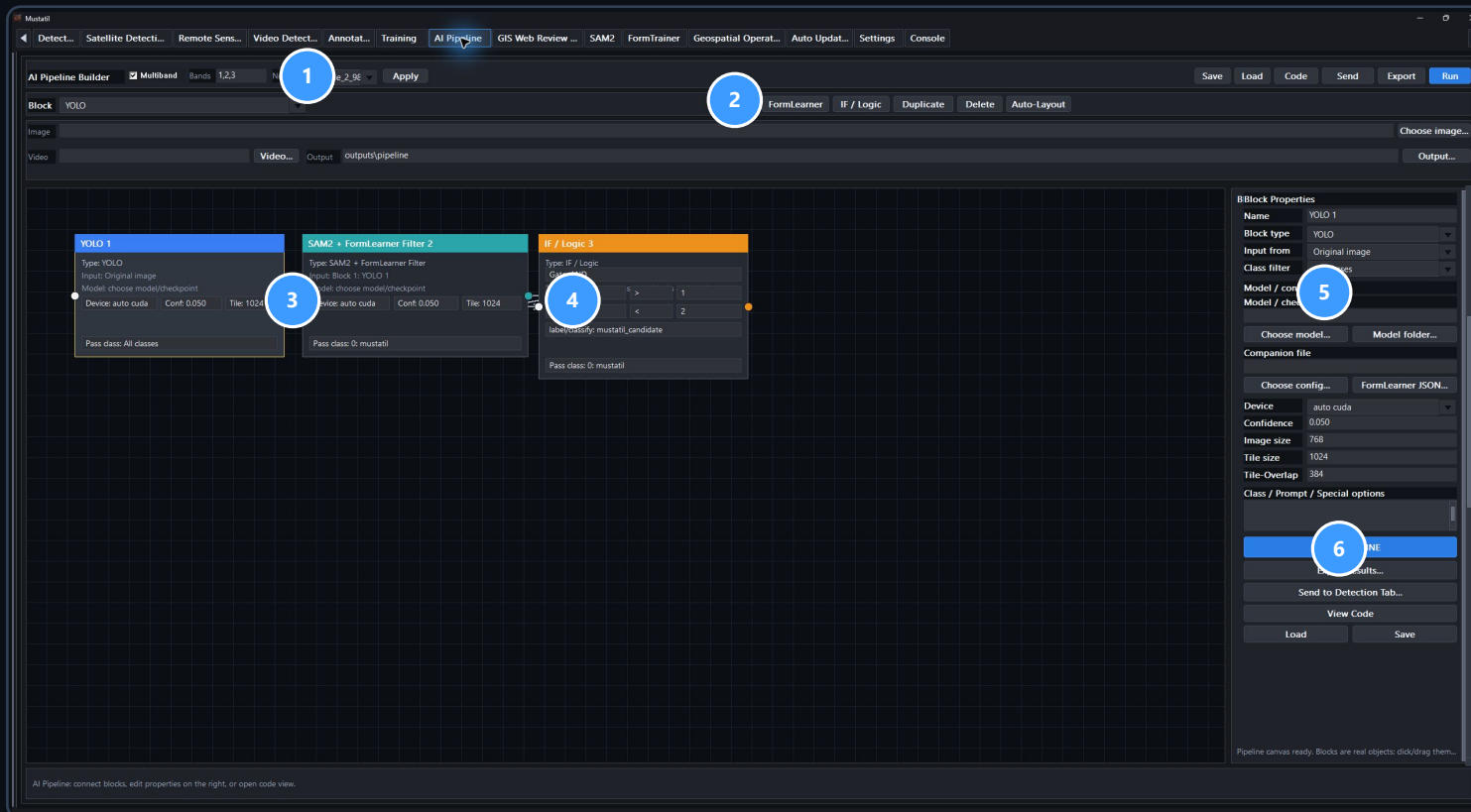
The screenshot displays the Custom Backbone Studio interface. On the left, there's a sidebar with various settings for training, including dataset, project, and training options. The main workspace shows a neural network architecture diagram with blocks like Stem Conv 1, Residual C2, Downsample C3, SE Attention, and SPP Block 5. Numbered callouts (1-6) highlight key actions: 1. Clicking 'Add' to add a block. 2. Dragging between colored ports to connect blocks. 3. Zooming and panning the graph. 4. Selecting a block to edit its properties. 5. Viewing the generated Python code. 6. Exporting the backbone to a .py file.

- 1 Choose a type under Block and click Add.
- 2 Connect outputs and inputs by dragging between the colored ports. Gate controls logical connections.
- 3 Zoom with the mouse wheel; pan the graph by left-dragging.
- 4 Select a block and edit Type, Channels, Stride, Repeat, Norm, and Activation on the right.
- 5 Use View Code to inspect the generated Python contract.
- 6 Export Python .py saves the backbone. Use custom backbone applies it to training.

Before training

Check channel counts, strides, and FPN outputs. A structurally invalid backbone may fail only during the first forward pass.

AI Pipeline: Connect Models and Logic



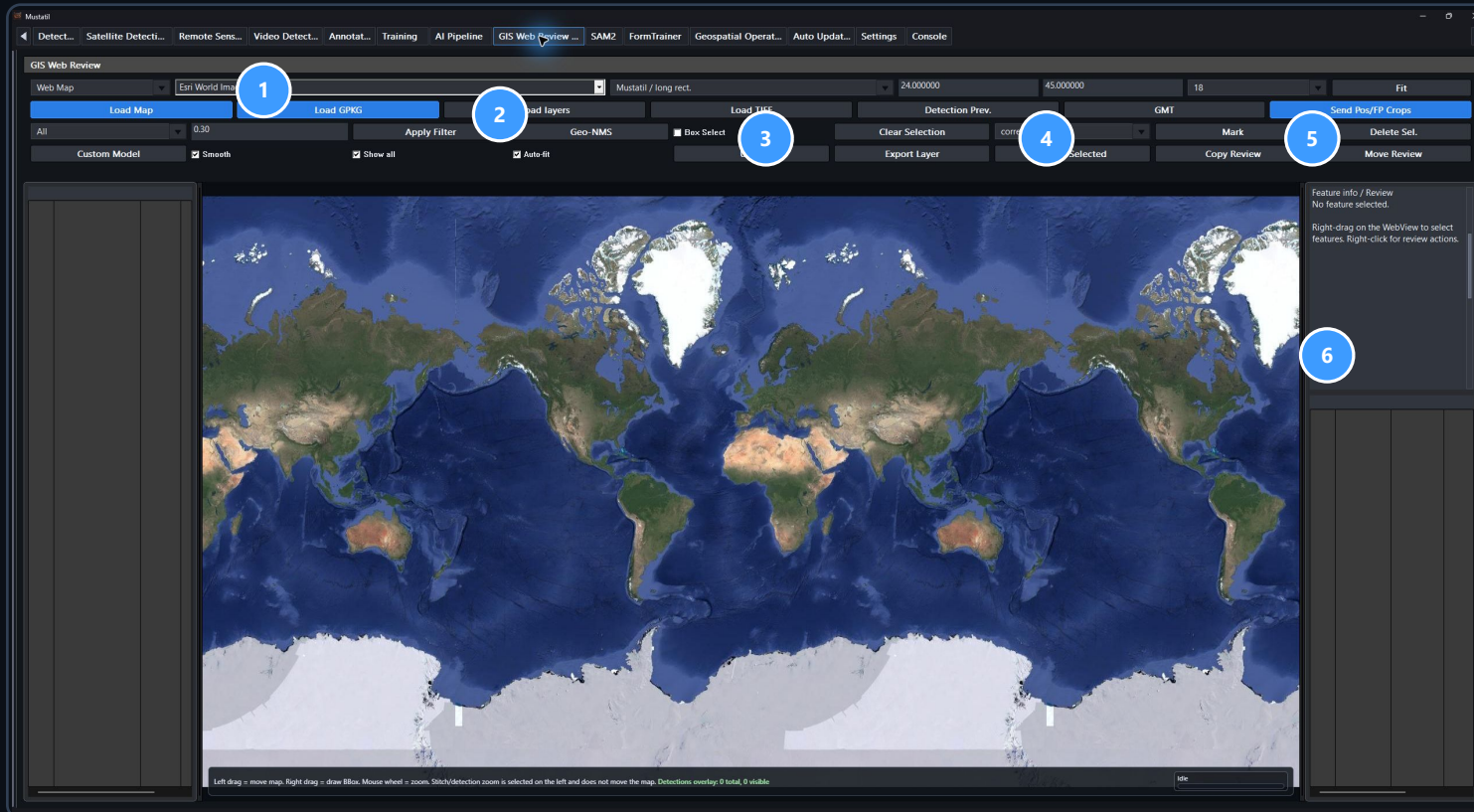
STEP BY STEP

- 1 Set Multiband, Bands, and Norm, then click Apply.
- 2 Choose Block, Add, FormLearner, or IF / Logic; use Duplicate and Delete to edit the graph.
- 3 Connect blocks by dragging their ports. Input from determines the data source.
- 4 Configure the model, class filter, Device, Confidence, Image size, Tile size, and Overlap.
- 5 At the top, choose Image or Video and select the output folder.
- 6 Start RUN PIPELINE. Export results saves the output; Send to Detection Tab transfers it to the map view.

Navigation

The mouse wheel zooms the pipeline graph. Left-drag to pan; block properties remain anchored to their block.

GIS Web Review: Inspect Results in Context



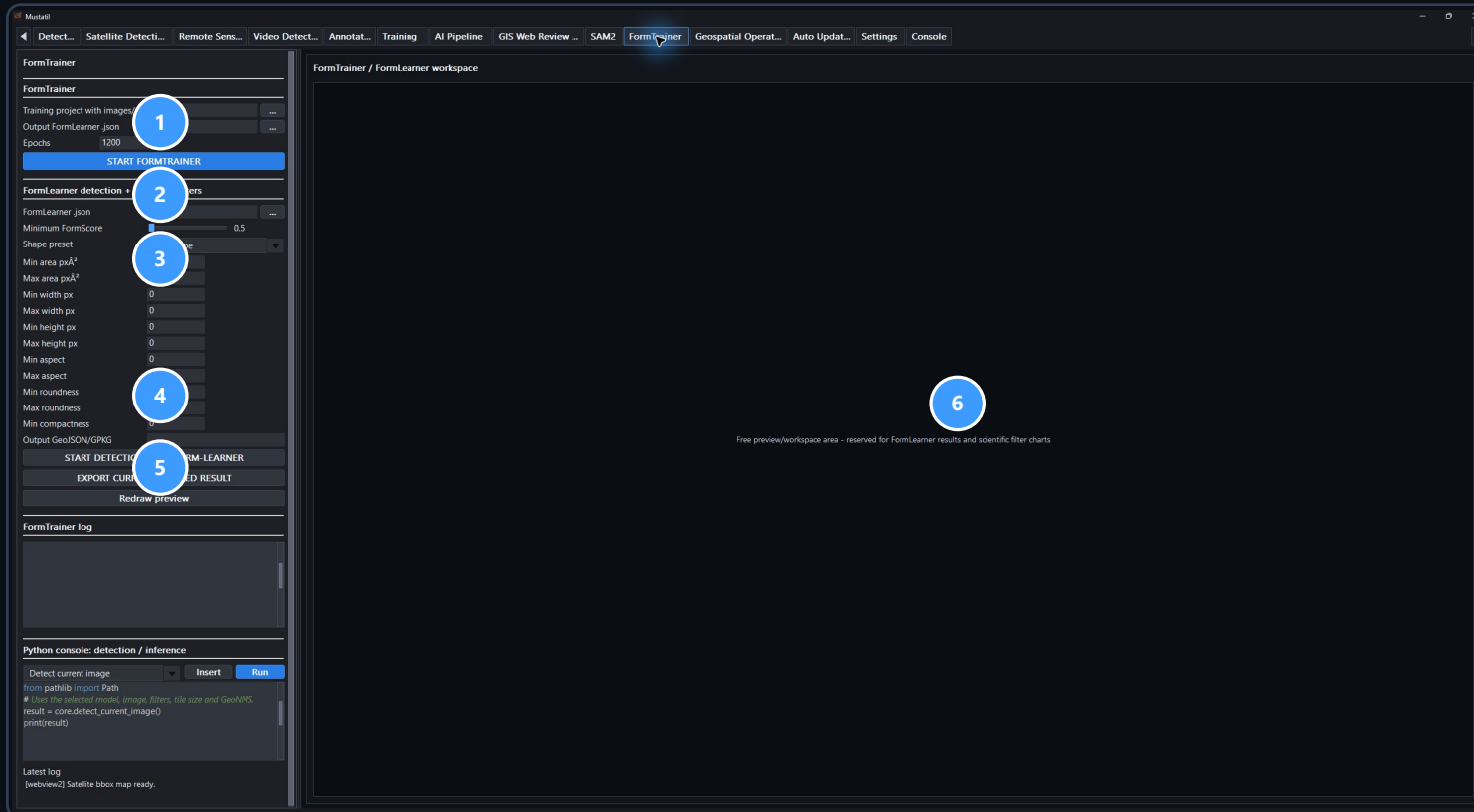
STEP BY STEP

- 1 Select Web Map and a background map; Load Map loads the basemap.
- 2 Load GPKG, Layers, or TIFF. Detection Prev. imports previous Detection results.
- 3 Filter by class and Confidence; Geo-NMS reduces overlapping detections.
- 4 Enable Box Select and right-drag to draw a selection area.
- 5 Mark detections, export the Layer/Selection, or copy/move them into a review dataset.
- 6 Review feature information on the right. Undo reverts the last review action.

Caution

Delete Sel. removes selected features. Export or back up the layer before larger review sessions.

FormTrainer and Scientific Filters



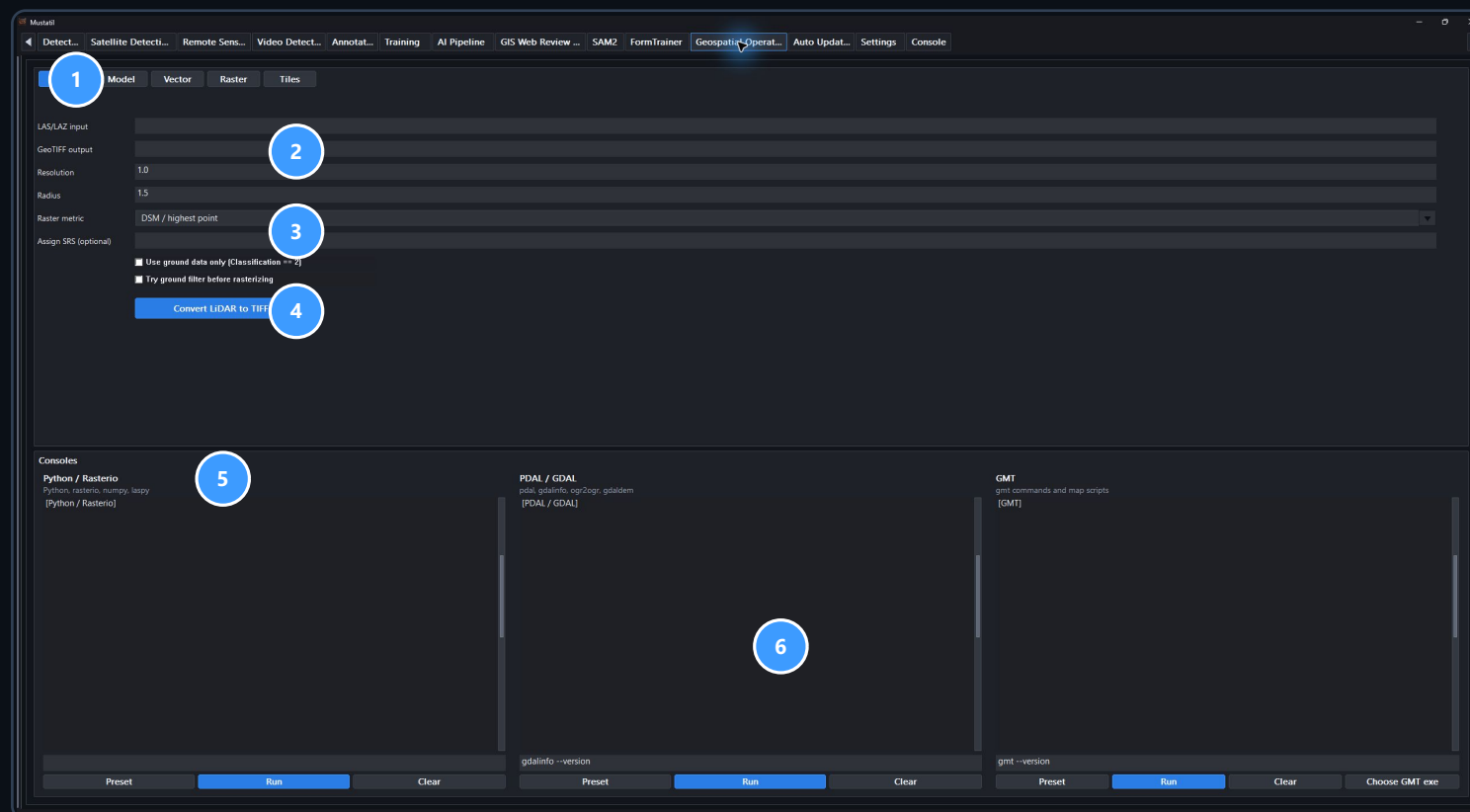
STEP BY STEP

- 1 Select the Training project containing images/annotations and an Output FormLearner .json.
- 2 Set Epochs and start START FORMTRAINER.
- 3 For inference, select the FormLearner .json and the Minimum FormScore.
- 4 Configure the Shape preset and limits for area, width, height, aspect ratio, roundness, and compactness.
- 5 Set Output GeoJSON/GPKG and start START DETECTION WITH FORM-LEARNER.
- 6 Review the preview and charts; EXPORT CURRENT FILTERED RESULT saves the filtered detections.

Filtering strategy

Begin with a few well-justified limits. Too many narrow filters can silently remove valid detections.

Geospatial Operations: LiDAR and Runtime Consoles



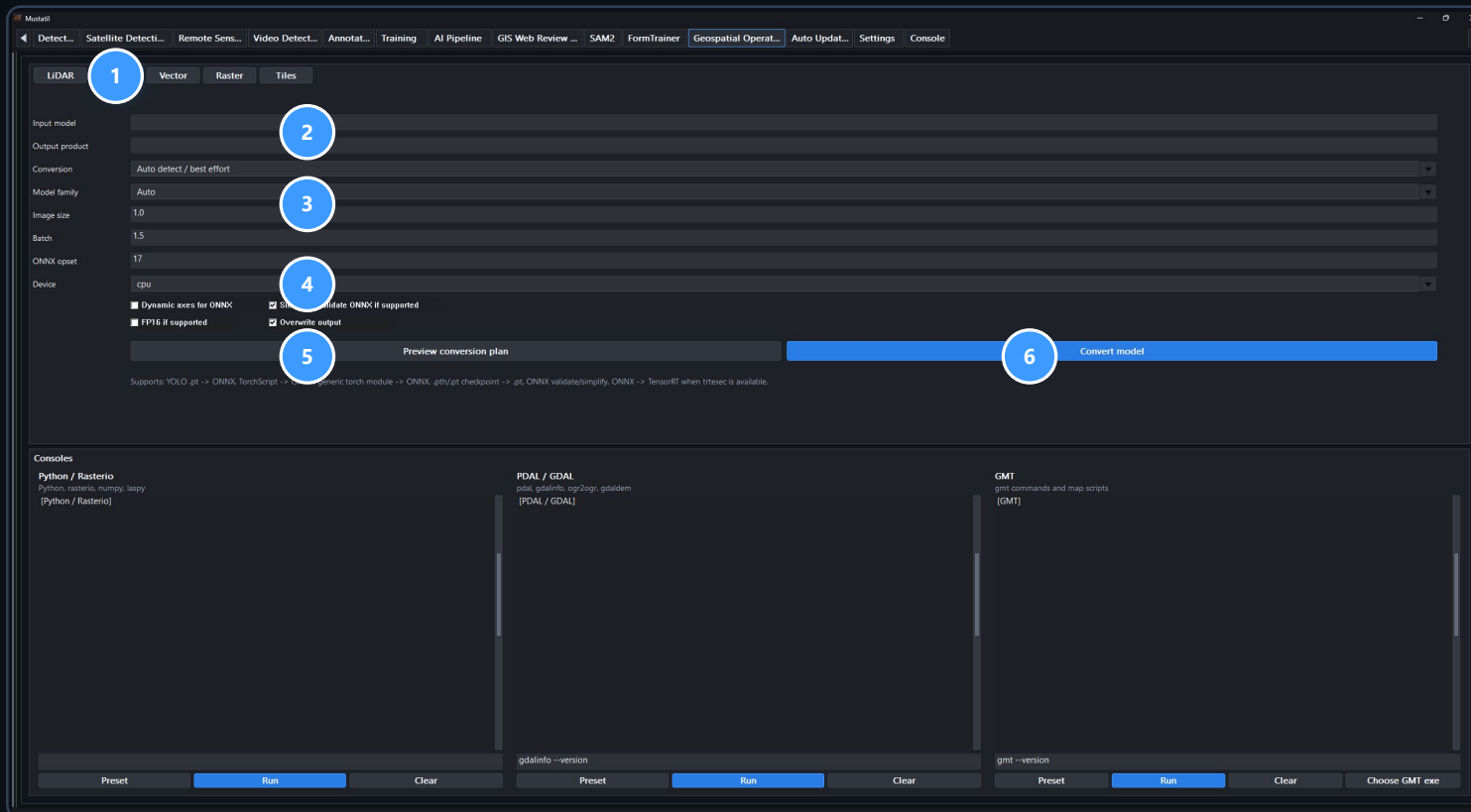
STEP BY STEP

- 1 Open LiDAR and select LAS/LAZ input and GeoTIFF output.
- 2 Set Resolution and Radius to match the point density.
- 3 Choose a Raster metric, such as DSM / highest point.
- 4 Optional: assign an SRS, use only Ground Classification == 2, or try the Ground filter.
- 5 Click Convert LiDAR to TIFF. The PDAL/GDAL console shows progress.
- 6 Separate consoles for Python/Rasterio, PDAL/GDAL, and GMT appear below with Preset, Run, and Clear.

CRS

Assign an SRS only when it is known with certainty. An incorrect CRS will displace all subsequent GIS results.

Geospatial Operations: Model Conversion



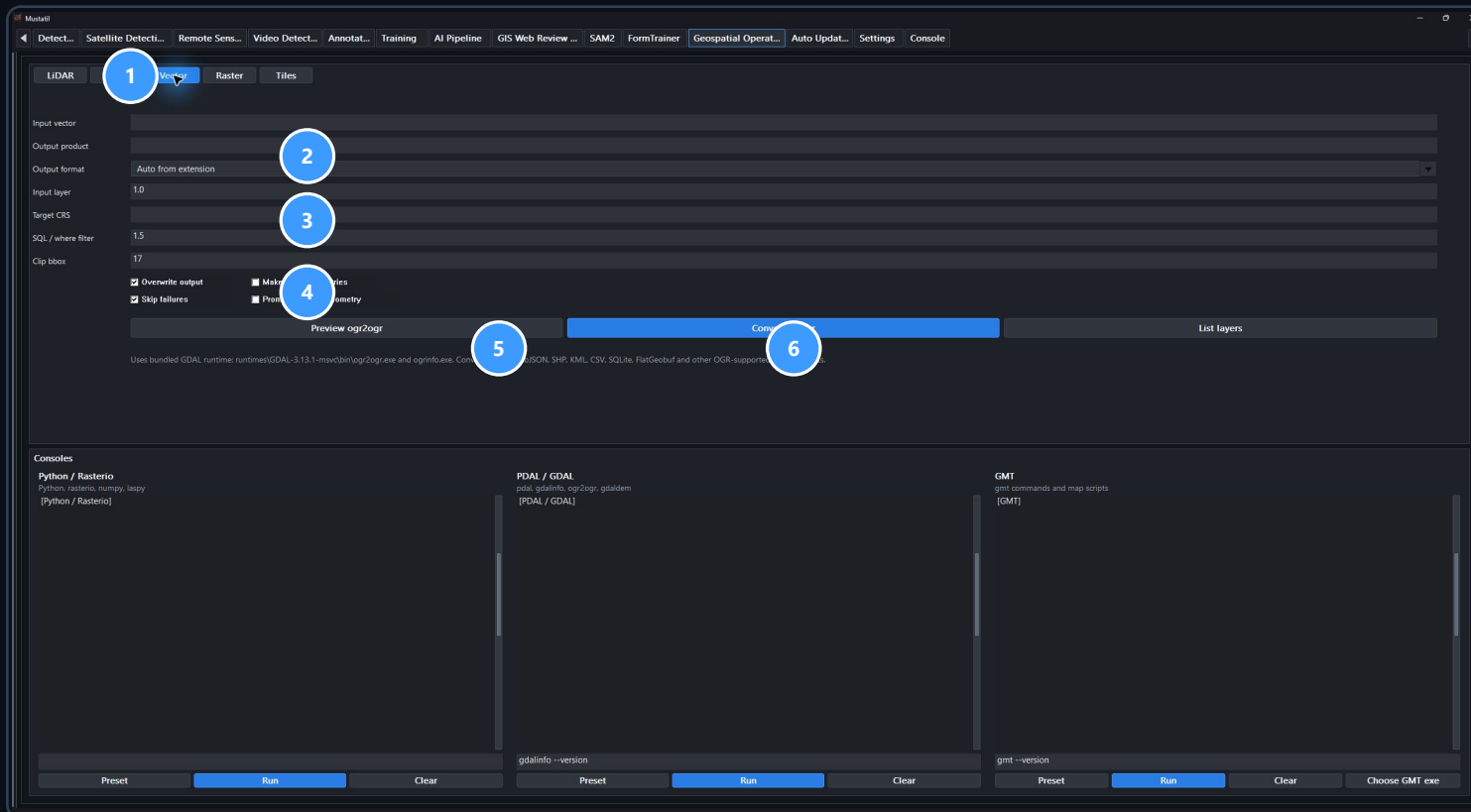
STEP BY STEP

- 1 Open Model and select the Input model and Output product.
- 2 Choose Conversion or leave Auto detect / best effort enabled.
- 3 Set Model family, Image size, Batch, ONNX opset, and Device.
- 4 Enable Dynamic axes, FP16, Simplify/validate, and Overwrite output as required.
- 5 Preview conversion plan shows the planned converter without creating output.
- 6 Convert model starts conversion, for example from YOLO .pt to ONNX.

Compatibility

FP16 and TensorRT require compatible GPU and runtime support. Validate ONNX before using the model in a pipeline.

Geospatial Operations: Vector Formats



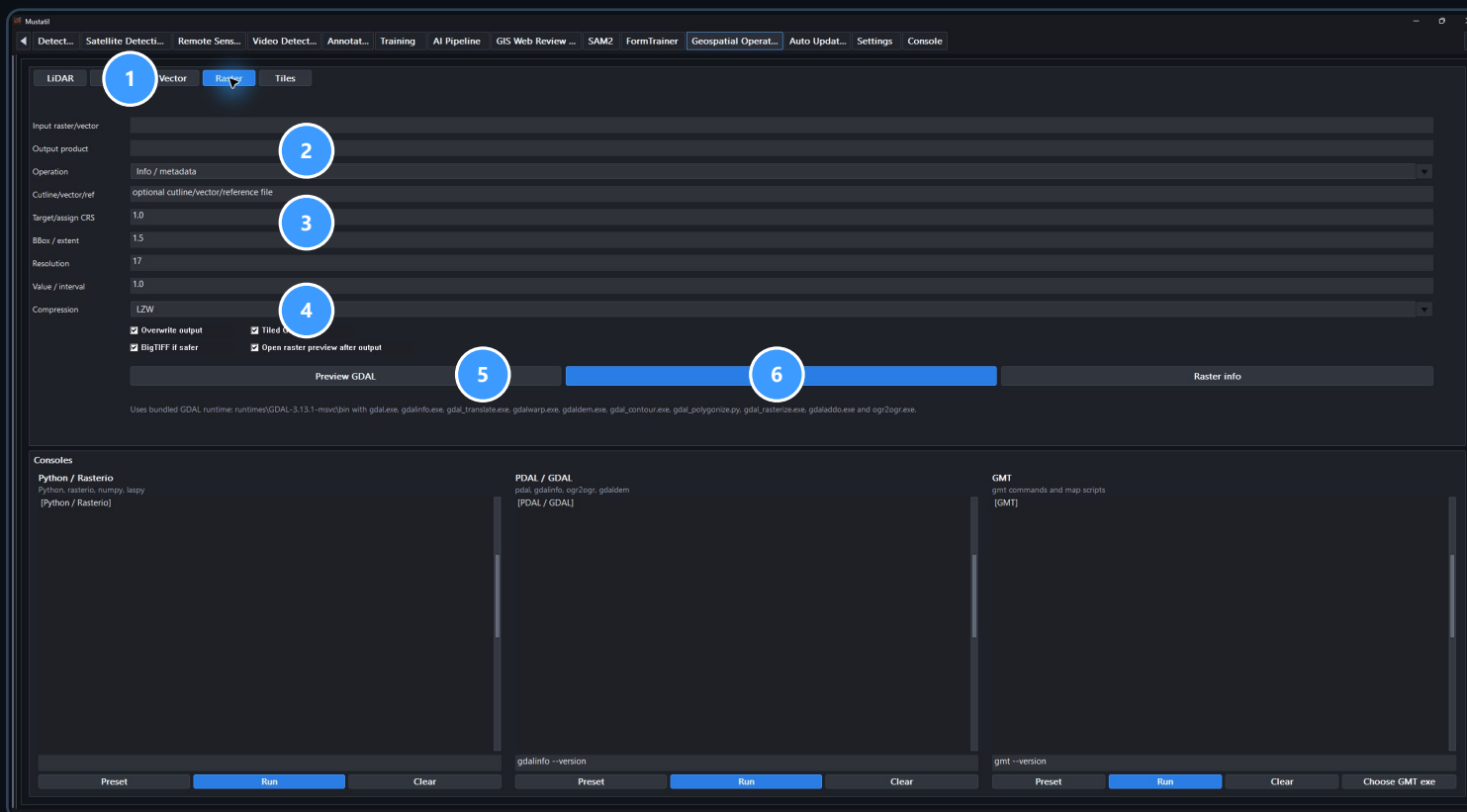
STEP BY STEP

- 1 Select the Input vector and Output product.
- 2 Choose the Output format or use Auto from extension.
- 3 Optional: enter the Input layer, Target CRS, SQL / where filter, and Clip bbox.
- 4 Choose Overwrite, Skip failures, Make valid geometries, or Promote to multi geometry.
- 5 Preview ogr2ogr shows the generated GDAL/OGR command.
- 6 Convert vector performs the conversion; List layers shows the input layer names.

Formats

Supported formats include GPKG, GeoJSON, SHP, KML, CSV, SQLite, and FlatGeobuf through the bundled GDAL/OGR runtime.

Geospatial Operations: Raster



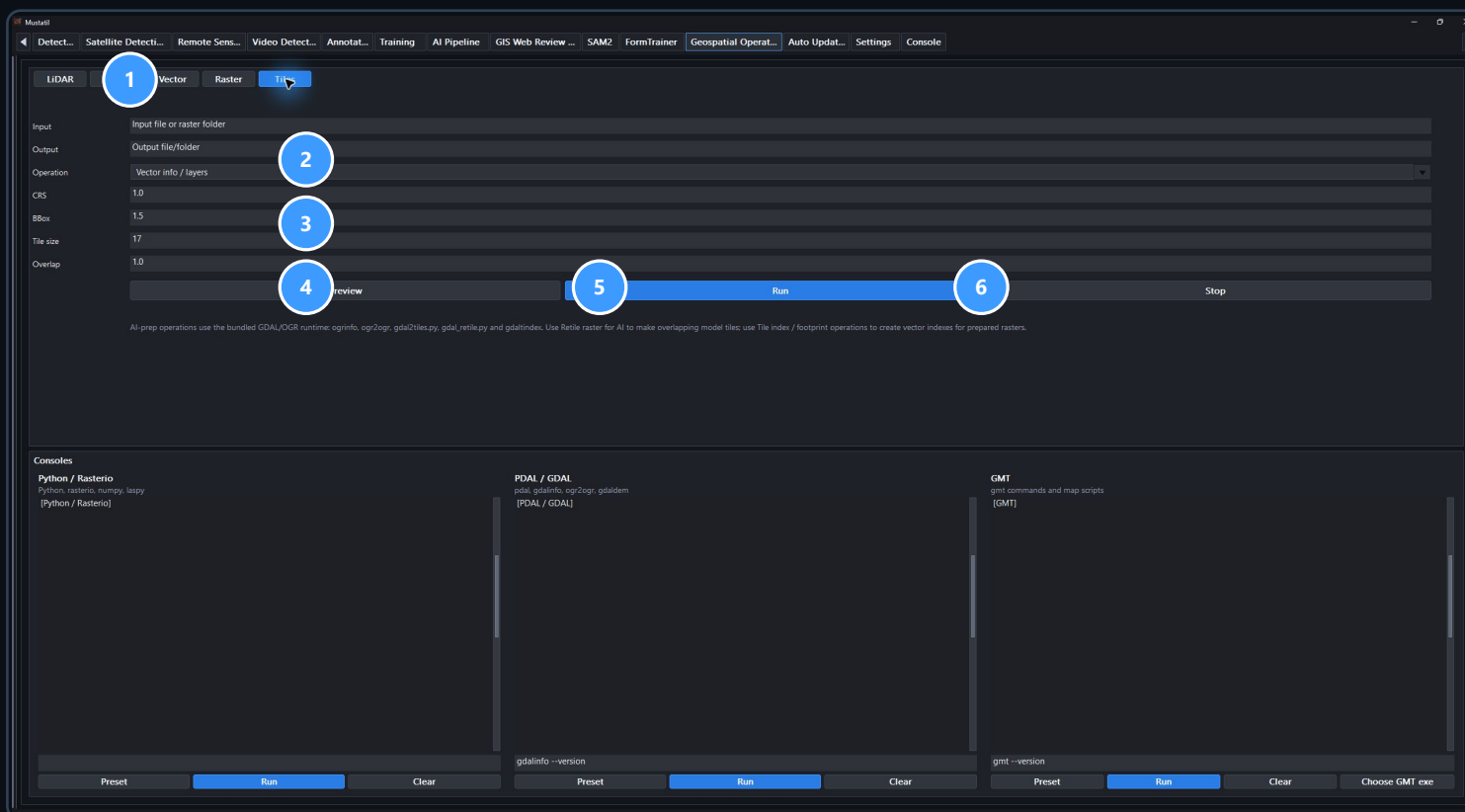
STEP BY STEP

1. Select the Input raster/vector and Output product.
2. Choose an Operation, such as Info / metadata, Translate, Warp, Clip, or Polygonize.
3. Set Cutline/reference, Target/assign CRS, and BBox / extent only when required.
4. Configure Resolution, Value / interval, and Compression.
5. Enable Tiled GeoTIFF, BigTIFF if safer, and Open raster preview after output.
6. Preview GDAL shows the planned command; Run raster starts it, and Raster info reads metadata.

BigTIFF

Keep BigTIFF if safer enabled for very large outputs. LZW provides lossless compression but requires additional processing time.

Geospatial Operations: Tiles and AI Preparation



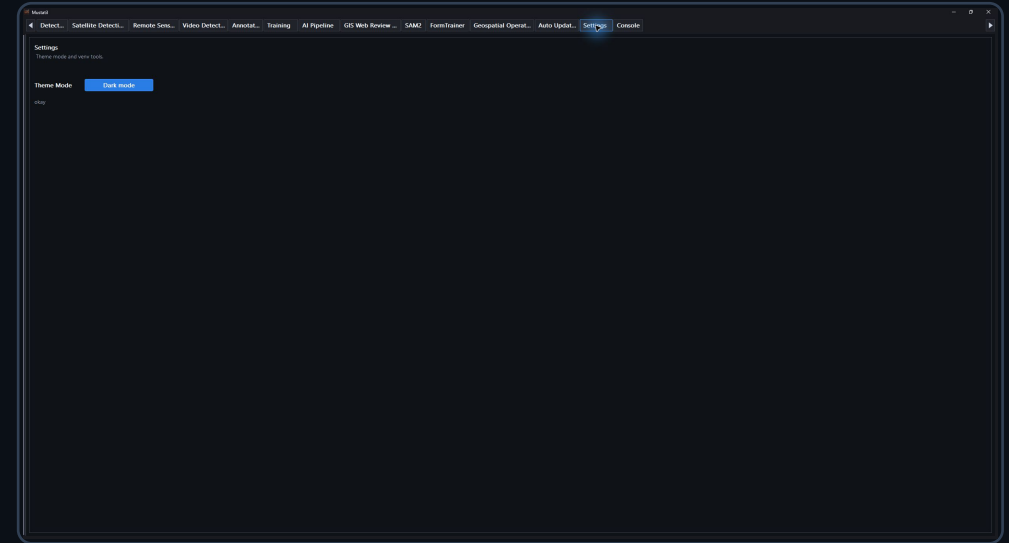
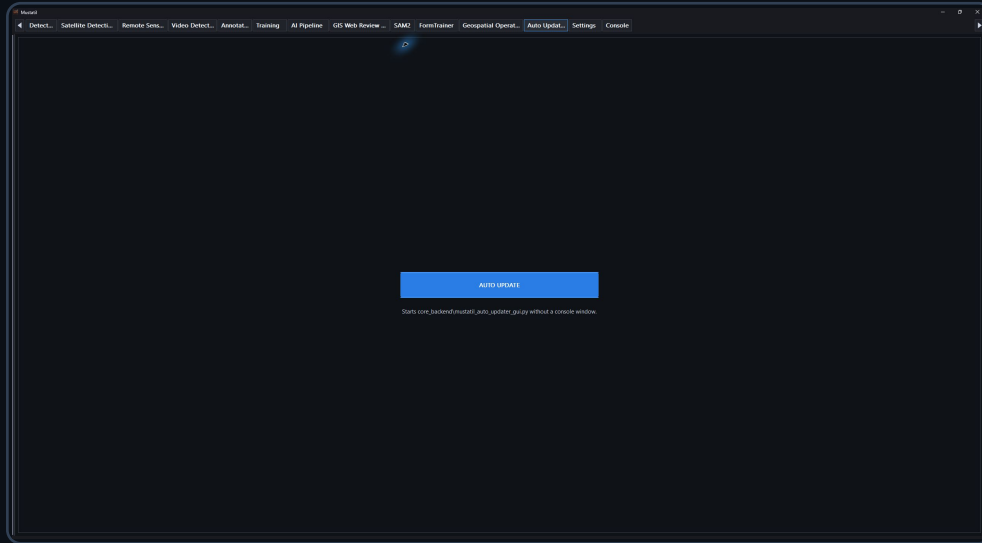
STEP BY STEP

- 1 Select the Input file/raster folder and Output file/folder.
- 2 Choose an Operation, such as Retile raster for AI or Tile index / footprint.
- 3 Set the CRS and optionally a BBox.
- 4 Set Tile size and Overlap. Overlap reduces edge losses during detection and training.
- 5 Preview shows the planned operation; Run starts it.
- 6 Stop ends an active tile job. Monitor progress in the GDAL console.

AI-Prep

Use the same Tile size and a similar Overlap as the later training or Detection run to keep data distributions consistent.

Auto Updater and Settings



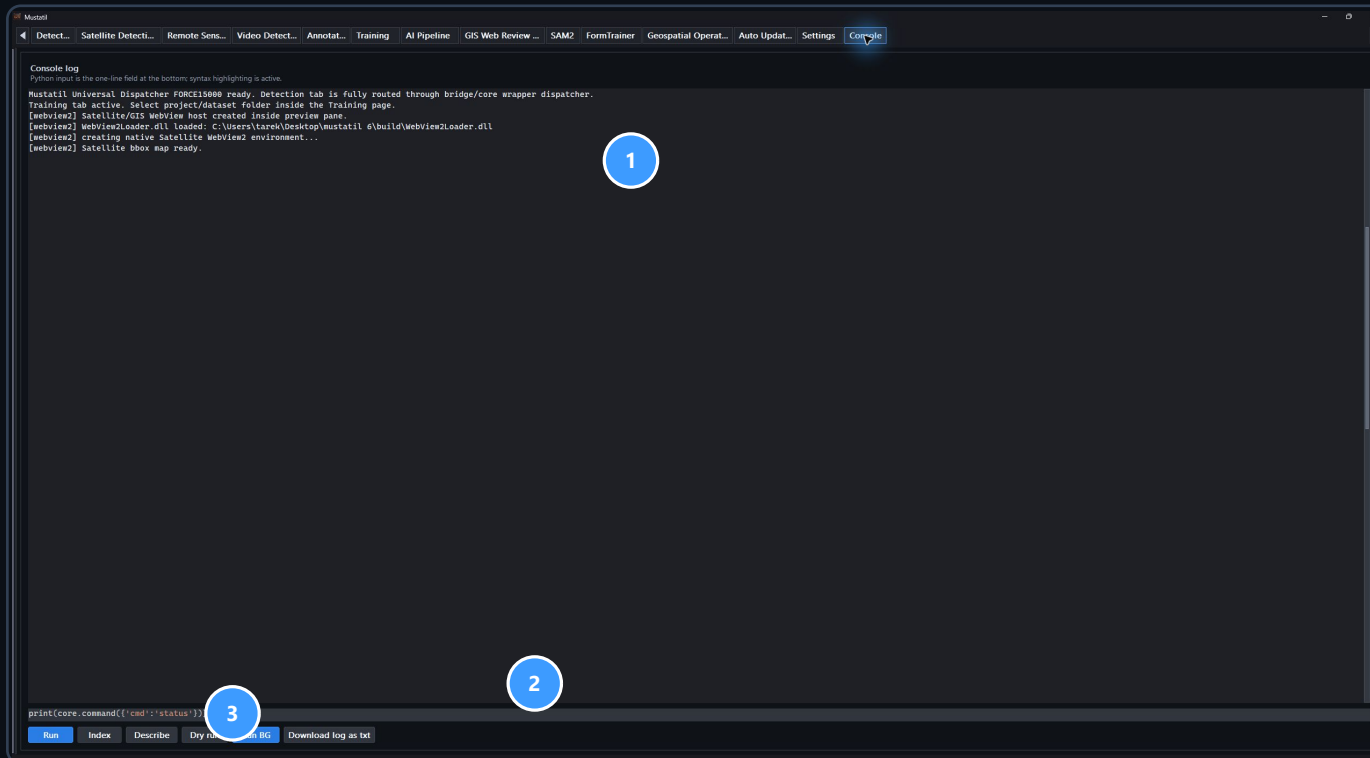
AUTO UPDATER

- 1 Open Auto Updater and click AUTO UPDATE.
- 2 The updater runs without a console window through the bundled venv and pythonw.exe.
- 3 Close Mustatil only when the updater asks you to do so.

SETTINGS

- 1 Open Settings.
- 2 Click Theme Mode to switch between Dark mode and Bright mode.
- 3 The selected mode applies to the entire interface.

Console and Troubleshooting



- 1 The Console log shows startup messages, runtime paths, and errors from all main areas.
- 2 The single-line input accepts Python/Core commands. Run executes the visible command.
- 3 Index and Describe explain available commands; Dry run checks without execution, while Run BG starts background work.
- 4 Download log as txt saves the log for sharing.

RUNTIME CHECKLIST

Python: build\.venv\Scripts\python.exe

Python without console: build\.venv\Scripts\pythonw.exe

GDAL: runtimes\GDAL-3.13.1-msvc\bin\gdal.exe

PDAL: runtimes\PDAL-2.10.2-install\pdal_launcher.exe

LAE-DINO: runtimes\LAE-DINO\python\python.exe

RF-DETR models: runtimes\RF-Detr\models

Four Proven End-to-End Workflows

A RASTER -> DETECTION -> GPKG

- 1 Open Detection and select the model and cuda.
- 2 Load the raster and choose Core or GDAL.
- 3 Set Tile/Overlap/GeoNMS and start Detection.
- 4 Review the boxes and export all detections as GPKG.

B SATELLITE -> BIGTIFF -> REVIEW

- 1 Open Satellite Detection and draw a BBox.
- 2 Load the BigTIFF or start Detection directly.
- 3 Send the results to GIS Web Review.
- 4 Filter, mark, and export the review layer.

C ANNOTATION -> TRAINING -> ONNX

- 1 Annotate images and save the labels.
- 2 Select the training project and output folder.
- 3 Dependency Check, Prepare Dataset, Train Model.
- 4 Review Metrics and export the final model to ONNX.

D PIPELINE -> FILTER -> GIS

- 1 Connect YOLO, SAM2/FormLearner, and IF/Logic.
- 2 Set models, classes, Device, and thresholds.
- 3 Run the pipeline and export the results.
- 4 Use Send to Detection Tab or GIS Review.